

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



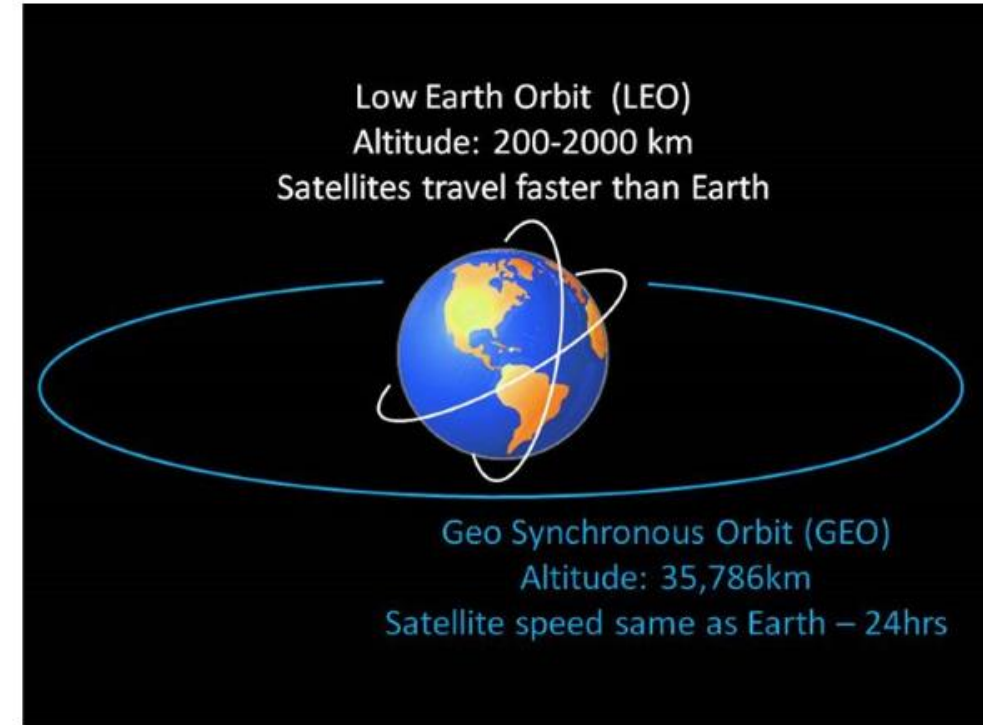
A COMPREHENSIVE COVERAGE

MODULE - 2

TYPES OF ORBITS - 2

LOW EARTH ORBIT (LEO)

- It is a range of orbits around the Earth with an altitude between 180km to 2000km.
- Any object below the minimum altitude (180km) will rapidly descend into the atmosphere, either burning up or crashing on the surface.
- LEO is the simplest and cheapest for satellite placement so a majority of satellites are placed in LEO. They move at extremely high speeds in LEO (~ 7 km/s).



The International Space Station (ISS) is in LEO at about 400 km from the Earth's surface. The Hubble Space Telescope orbits at about 540 km.

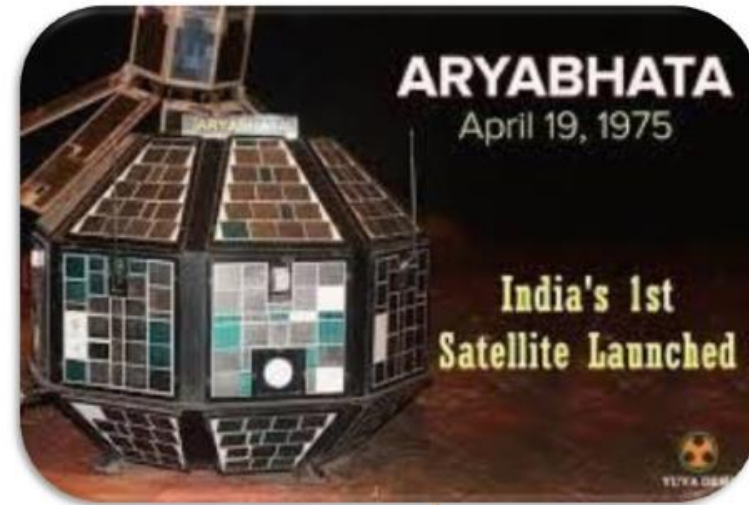
COMMUNICATION SATELLITES IN LEO

- 1** Communication satellites require geostationary orbit to provide continuous transmission and LEOs are not geostationary.
- 2** But it requires less energy to place a satellite into LEO and also satellites need less powerful amplifiers to transmit signal, so many communication satellites are placed in LEO.
- 3** Satellites in LEO have a small field of view, it means they are only able to observe and communicate with a small area on the Earth at a time. So, a network or a Constellation of satellites (a group of satellites) is required.
- 4** As a result of this the LEO environment is getting congested with space debris.



ARYABHATA

- Aryabhata is the first satellite built by India.
- It was named after the prominent Indian astronomer and mathematician of the 5th century CE.
- It was placed in LEO.
- The satellite was assembled in India, but was launched from within the Soviet Union by a Russian-made rocket on April 19, 1975.
- It contained instruments to explore Earth's ionosphere, measure neutrons and gamma rays from the Sun.



INTERNATIONAL SPACE STATION

- The International Space Station (ISS) is a habitable artificial satellite, in Low Earth Orbit.
- Its first component launched into orbit in 1998 and the station is expected to operate until 2028. Development and assembly of the station continues.
- The ISS is the largest human-made body in orbit and can often be seen with the naked eye from Earth.
- The ISS serves as a microgravity and space environment research laboratory in which crew members conduct experiments in human biology, physics, astronomy, weather study etc.



- The ISS maintains an orbit with an altitude of between 330 and 435 km. It completes 15.5 orbits per day.
- The ISS programme is a joint project among five participating space agencies:
 - ✓ NASA
 - ✓ Roscosmos (Russian Space Agency)
 - ✓ JAXA (Japanese Space Agency)
 - ✓ ESA (European Space Agency) and
 - ✓ CSA (Canadian Space Agency).
- The ownership and use of the space station is established by intergovernmental treaties and agreements between these countries.
- The station is divided into two sections, the Russian Orbital Segment (ROS) and the United States Orbital Segment (USOS), which is shared by many nations.



INTERNATIONAL SPACE STATION

- The Russian Orbital Segment is the name given to the components of the International Space Station (ISS) constructed in Russia and operated by Roscosmos. The ROS handles Guidance, Navigation & Control for the entire Station.
- United States Orbital Segment is the name given to the components of the International Space Station (ISS) constructed and operated by -
 - ✓ National Aeronautics and Space Administration (NASA)
 - ✓ European Space Agency (ESA)
 - ✓ Canadian Space Agency (CSA) and
 - ✓ Japan Aerospace Exploration Agency (JAXA).
- The station has been continuously occupied for about 20 years since November 2000. This is the longest continuous human presence in low Earth orbit.

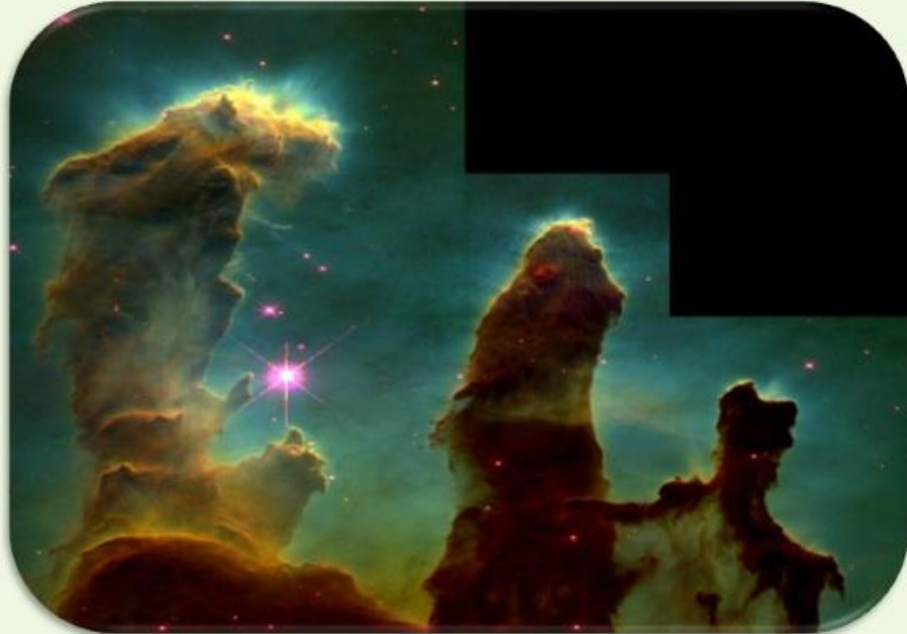


HUBBLE SPACE TELESCOPE

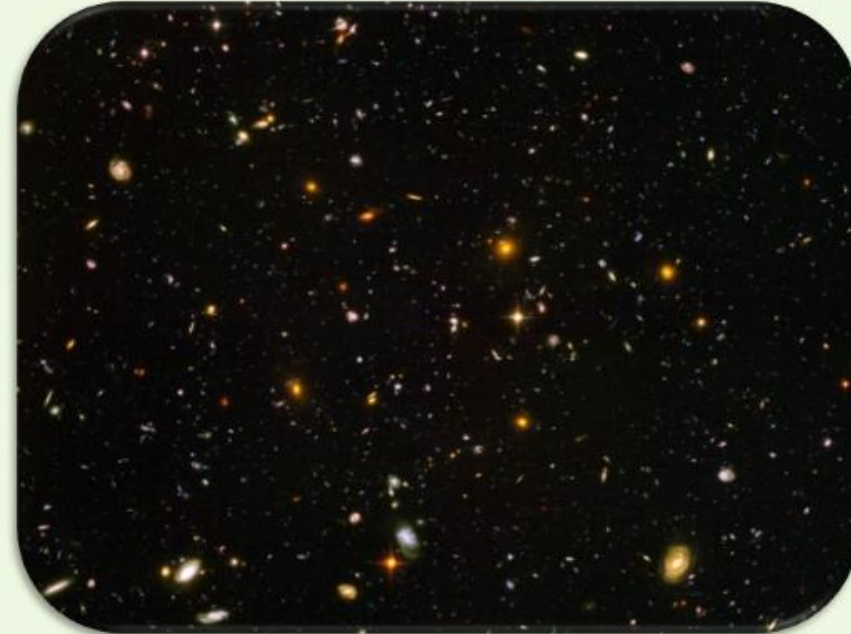
- The Hubble Space Telescope (HST) is a space telescope that was launched into low Earth orbit in 1990 and remains in operation. The HST is named after the astronomer Edwin Hubble.
- Hubble has a 2.4-meter (7.9 ft) mirror, Hubble can observe in the near ultraviolet, visible, and near infrared spectra.
- Hubble's orbit outside the distortion of Earth's atmosphere allows it to take extremely high-resolution images.
- Hubble has recorded some of the most detailed images ever, allowing a deep view into space and time. Many Hubble observations have led to breakthroughs in astrophysics.



HUBBLE SPACE TELESCOPE



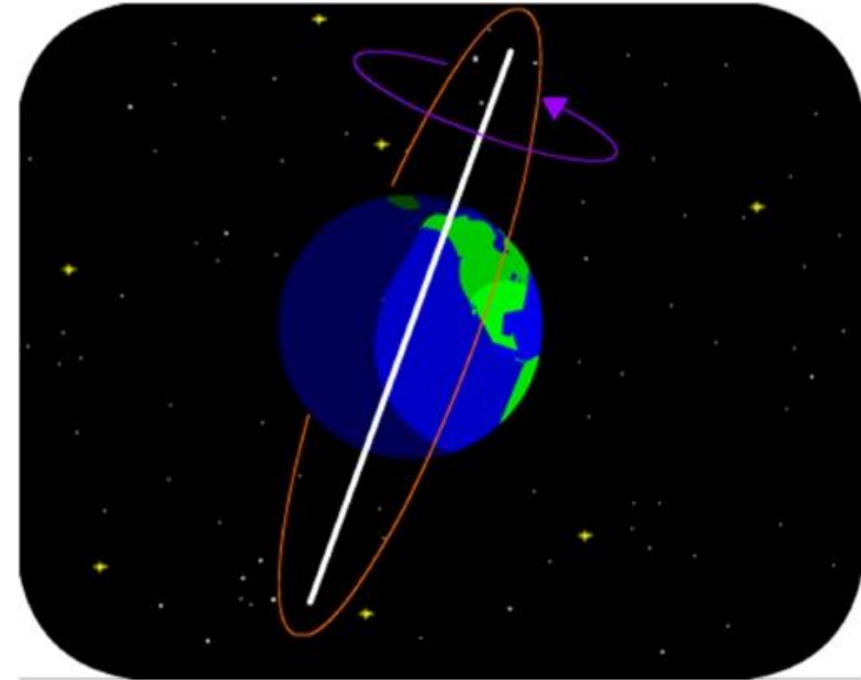
**IMAGE OF EAGLE NEBULA
TAKEN BY HST**



**THE HUBBLE DEEP FIELD -
ONE OF THE GREATEST
ACHIEVEMENTS OF HST**

POLAR ORBIT

- A satellite in polar orbit travels north-south over the poles and takes approximately an hour and a half for a full revolution.
- As the satellite is in orbit, the Earth is rotating beneath it. As a result, a satellite can observe the entire Earth's surface in the time span of 24 hours. So polar orbit is used for earth mapping and earth observing satellites.
- A low altitude polar orbit is widely used for monitoring the Earth.
- The orbital path of the satellite does not have to cross the poles exactly for an orbit to be called polar, an orbit which passes within 20 to 30 degrees of the poles is still a polar orbit.



WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

0866 244 44 72

98499 42299

94415 27588